

自定义函数

人员

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上周作业检查

上周作业链接: <https://cppoj.kids123code.com/contest/4811>

🏠 比赛概况

📋 题目列表

📋 选择题列表

📄 提交记录

★ 实时榜单

王向东老师周六三点半C++字符串练习4

🔄 刷新

#	用户名	姓名	编程分	时间	A	B	C	D	E	F	G	H
1	zhangjiarui	张嘉芮	800	78	100	100	100	100	100	100	100	100
2	xiejuntian	解钧添	800	78	100	100	100	100	100	100	100	100
3	lihonghao	李弘浩	800	79	100	100	100	100	100	100	100	100
4	sujiacun	苏佳骏	800	80	100	100	100	100	100	100	100	100
5	quyuxuan	屈钰轩	800	83	100	100	100	100	100	100	100	100
6	chenyiqi	陈奕圻	800	83	100	100	100	100	100	100	100	100
7	wangchunshuo	王纯硕	700	79	100	100	100	100	100	100	100	
8	zhaotingshu	赵庭澍	700	79	100	100	100	100	100	100	100	0
9	pengjunwen	彭俊文	700	81	100	100	100	100	100	100	100	

作业

<https://cppoj.kids123code.com/contest/4951> (作业是 A ~ H 题必做)

课堂表现

今天课上讲了 自定义函数, 同学们课上掌握不错, 课下也要多复习一下, 不然容易遗忘。

课堂内容

判断闰年

```
#include <bits/stdc++.h>

using namespace std;

bool check(int x) {
    if (x%400==0 || (x%4==0&& x%100!=0)) {
        return true;
    }
    else {
        return false;
    }
}
```

```
int main() {
    int n;
    cin >> n;
    if (check(n)) {
        cout << "Y";
    }
    else {
        cout << "N";
    }
    return 0;
}
```

质数判断进阶版

```
#include <bits/stdc++.h>

using namespace std;

bool check(long long x) {
    if (x <= 1) return false;
    for (long long i = 2; i*i <= x; i++) {
        if (x%i == 0) {
            return false;
        }
    }
    return true;
}

int main() {
    long long n;
    cin >> n;
    if (check(n)) {
        cout << "Yes";
    }
    else {
        cout << "No";
    }
    return 0;
}
```

编程求 $1 * 2 * 3 * \dots * n$

```
#include <bits/stdc++.h>

using namespace std;

int f(int x) {
```

```
int res = 1;
for (int i = 1; i <= x; i++) {
    res *= i;
}
return res;
}

int main() {
    int n;
    cin >> n;
    cout << f(n);
    return 0;
}
```

从大到小排序

```
#include <bits/stdc++.h>

using namespace std;

int a[1000005];
bool cmp(int x, int y) {
    return x > y;
}

int main() {
    int n;
    cin >> n;
    for (int i = 1; i <= n; i++) {
        cin >> a[i];
    }
    sort(a+1, a+n+1, cmp);
    for (int i = 1; i <= n; i++) {
        cout << a[i] << " ";
    }
    return 0;
}
```

整数奇偶排序

```
#include <bits/stdc++.h>

using namespace std;

int a[1000005];
bool cmp(int x, int y) {
    if (x%2==1 && y%2==1) {
        return x > y;
    }
}
```

```
}
else if (x%2==1 && y%2==0) {
    return true;
}
else if (x%2==0 && y%2==1) {
    return false;
}
else {
    return x < y;
}
}

int main() {
    int n = 10;
    for (int i = 1; i <= n; i++) {
        cin >> a[i];
    }
    sort(a+1, a+n+1, cmp);
    for (int i = 1; i <= n; i++) {
        cout << a[i] << " ";
    }
    return 0;
}
```

不一样的排序

```
#include <bits/stdc++.h>

using namespace std;

int a[1000005];
bool cmp(int x, int y) {
    int a = x, b = y, sum1 = 0, sum2 = 0;
    while (x != 0) {
        sum1 += x%10;
        x /= 10;
    }
    while (y != 0) {
        sum2 += y%10;
        y /= 10;
    }
    if (sum1 != sum2) {
        return sum1 < sum2;
    }
    else {
        return a < b;
    }
}

int main() {
    int n;
```

```
cin >> n;
for (int i = 1; i <= n; i++) {
    cin >> a[i];
}
sort(a+1, a+n+1, cmp);
for (int i = 1; i <= n; i++) {
    cout << a[i] << " ";
}
return 0;
}
```

计算2的幂

```
#include <bits/stdc++.h>

using namespace std;

int han(int a) {
    int res = 1;
    for (int i = 1; i <= a; i++) {
        res *= 2;
    }
    return res;
}

int main()
{
    int n;
    cin >> n;
    cout << han(n) << endl;
    return 0;
}
```

求三个数的最大数

```
#include <bits/stdc++.h>

using namespace std;

int han(int a, int b, int c) {
    return max(a, max(b, c));
}

int main()
{
    int a, b, c;
    cin >> a >> b >> c;
    cout << han(a, b, c) << endl;
}
```

```
    return 0;  
}
```